

basic induction, exposure to the production and services the organisation is involved in, hands-on designing skills with the CAD/CAM tools used, exposure to the organisation's business culture, and project-specific orientation. As the person goes higher on the rung, he is trained to do high-end designing work for products and structures.

Industry-Academy Collaboration

Academic institutions need better exposure association with the industries employing their engineers. Education and training must benefit the industry and the students as well. The students will thus be able to receive better industry exposure through summer-winter internships offered by many companies.

Organisations using CAD/CAM exclusively in their business model are now looking to join hands with academies that offer education and specialised training in CAD/CAM. Initiatives such as internships, campus placements, summer-winter projects, training programmes, and sponsoring R&D projects have now started, due to joint efforts by the industry and academies.

CAD/CAM software vendors provide consulting and guidance to the educational bodies on how to establish and maintain computing labs and keep them up-to-date with the latest software.

CAD software vendors such as Autodesk, SolidWorks, and UGS Corp. have their own software-specific courseware and certifications. Information about authorised training centres and resellers can be found at the Web sites of CAD software vendors.

Roles

People in different industries are given different designation during their tenure with an organisation. However, the nature of the work remains the same at each level.

Entry Level

Trainee Designer/Engineer: The job here is to do everything right from documentation of a project design to the actual design using CAD tools. At this stage the individual gets the maximum time to get his hands on the CAD software tool on a platform (Windows, Linux, or UNIX). Drafting, solid modelling, surface modelling, finishing, and all the other basic tasks involved in designing a new product, constructs and walkthroughs are among the main activities that are done.

During this tenure, the individual gets exposed to all areas of the project. One can expect to start with remuneration as low as Rs 1 lakh to 3 lakh per annum, depending upon organisation and industry. It takes roughly one to three years to advance to the next level.

Senior Designer/Engineer: Here, one validates the work of the Trainee Designer or Engineer, and works in one's own area of specialisation. People at this level are responsible for development of key areas as well as the growth of the trainee engineers. Responsibilities include the validating and



“People who have the skills to work on high-end designing projects are required to possess the respective domain knowledge. That helps them in understand how the product will eventually evolve from the raw material through production.”

Anil Sondur
Business Unit Head,
Innovation Design
Engineering,
TATA Elxsi

analytical part of the designs. The Senior Designer or Engineer reports to the Design Lead or Engineering Lead respectively. Remuneration is between Rs 3 to 5 lakh per annum. It takes experience of three to five years to advance to the next level.

Middle Level

Design Lead, or Lead Designer or Engineer: As a person reaches this level, the obligations and nature of work incline towards the management side. A Design or Engineering Lead is responsible for the team of designers and engineers under him. He works on a particular project and keeps a check on its progress. All the changes or finishing work required is under his influence. At this level, remuneration is between Rs 5 and 7 lakh per annum. The time needed to advance to the next level is, again, three to five years.

Project Manager or Project Lead (Design or Engineering): Responsible for the entire project. He interacts with clients and understands their needs and suggested changes. Orders regarding changes are passed on to the Design Lead. He manages the application part of a particular CAD technology or tool for a project, and keeps a check over the project performance at every level. Remuneration ranges from Rs 7 to 10 lakh per annum, and advancement to the next level takes four to six years.

Top Level

Design / Engineering Director: An individual at this level is involved in taking strategic and managerial decisions upon a project. The nature of work here is mostly managerial. The decision of selecting a CAD technology and the tools to be used for the projects is taken by the Design/Engineering Director. Remuneration would be more than Rs 10 lakh per annum.

A Final Word

According to a study by Booz Allen Hamilton and NASSCOM, engineering services could help to create 1.5 to 2 lakh additional jobs by 2020—up to 1 lakh with the multiplier effect. The global spending in engineering services will grow from \$750 billion to \$1 trillion by 2020, and India will raise its market share from 12 per cent in 2004 to 30 per cent by 2020. The total offshore engineering services are expected to grow from \$150 to \$225 billion, out of which 25 per cent will be realised by the potential talent pool and existing work force in engineering services.

Demand for CAD/CAM professionals who have specialised knowledge of a domain will grow. To breed them, Indian industries need to invest in infrastructure, academics, and the workforce now to reap the benefits by 2020.

Be it the consumer electronics, aeronautical, automotive, or any other industry where CAD/CAM/CAE tools are used, the demand for qualified professionals will always be on the rise. The return over investment in this career option are rewarding, since the offshoring of engineering services is already picking up.■

samir_makwana@thinkdigit.com